GridShield AI — From Zero to Hero

E.ON Innovation Challenge 2026 — Defensible Technical Architecture, Cluster 5 Adoption & Onboarding Guide

Team GridShield:	Target Infrastructure:	Technical Framework:
Pulkit Agrawal (Lead AI & Red Team) Kabir Roy (Cybersecurity & Purple SOC)	E.ON Power Grids, SCADA, OCPP EV Gateways, 9.6M Customer Assets	React 18 + Vite (Demo UI), PyTorch GraphSAGE + TinyML (Target Arch)

1. Executive Summary & Defensible Problem Statement

Modern energy distribution grids face a critical vulnerability: the rapid decentralisation of energy assets. Over 9.6 million customer-owned Distributed Energy Resources (DERs)—including EV chargers, solar inverters, and heat pumps—are connected to E.ON's distribution network across Europe. Coordinated cyberattacks targeting thousands of DERs simultaneously pose severe grid frequency perturbation and blackout risks under NIS2 regulatory frameworks.

GridShield AI addresses this challenge via an autonomous **Purple Team Security Orchestration Platform**. It continuously pairs an autonomous **Red Team Engine** (simulating MITRE ATT&CK; vectors) with a **Blue Team Anomaly Detector** (Edge TinyML + Cloud Graph Neural Network), delivering real-time threat detection and automated SOAR zero-trust isolation.

2. Technical Transparency: Prototype Demo vs Enterprise Roadmap

To ensure technical defensibility before Infosys & E.ON judges, GridShield AI explicitly demarcates what was built for the hackathon interactive demo versus the proposed production architecture:

Dimension	Built Hackathon Prototype (Interactive Demo)	Proposed Enterprise Production Architecture
User Interface	React 18 + Vite SPA with HTML5 Canvas Space Engine & Lucide Icons	Production SOC Web Application (React + D3.js + WebGL Canvas)
Simulation Engine	Scripted client-side simulation (attackEngine.js & assetMonitor.js)	PyTorch GraphSAGE GNN Model + Apache Kafka 4.2 GB/s Stream
Edge Execution	JavaScript web worker mock running anomaly scoring math	TinyML / TFLite Micro C++ binary (<800KB) on ARM Cortex-M4
Grid Topologies	9.6M virtual customer DER nodes across 5 device categories	Neo4j Graph Database mapping substation links & SCADA gateways

3. Itemized Techno-Economic Cost Model (€0.80 / Device / Year)

Per E.ON's evaluation rubric requiring transparent financial assumptions, the headline €0.80/device/year cost at 500K scale is broken down as follows:

Component	Cost / Device / Year	Techno-Economic Assumption & Work Shown
1. Edge MCU Execution	€0.12	Software overlay on existing device MCU. Zero new hardware cost (<800KB Flash, 4KB RAM).
2. Telemetry Ingestion	€0.28	Compressed MQTT telemetry (<2KB/hr/device) ingested into multi-tenant Apache Kafka bus.
3. GraphSAGE GNN Compute	€0.22	Subsampled PyTorch GNN graph embedding on shared Kubernetes GPU cluster (NVIDIA T4).
4. Maintenance & Audit	€0.18	Automated NIS2 Article 21 compliance reporting & signed ED25519 OTA firmware updates.
TOTAL TCO	**€0.80 / dev / yr**	**ROI positive in Year 1; saves €12.6M annually per DSO in avoided breach & NIS2 penalty costs.**

4. Cluster 5 Customer Adoption & Incentive Framework

Addressing Cluster 5's scored requirement ('How to convince asset owners to connect devices to central monitoring'), GridShield AI introduces 4 incentive pillars:

■ 4–8% Dynamic Grid Tariff Discount

DSOs grant direct monthly electricity bill credits to customers who opt into GridShield's TinyML edge telemetry network.

■ Up to 25% Cyber Insurance Rebate

Underwriting partners (Allianz, AXA) lower insurance premiums for home batteries & EV chargers running verified zero-trust firmware.

■ 100% Zero-Privacy Exposure Guarantee

Federated TinyML runs locally on device MCUs — raw household usage data never leaves the home, satisfying strict GDPR demands.

■■ Hardware Exploitation Replacement Guarantee

DSOs cover full zero-cost replacement for any DER asset damaged by grid frequency surges or unauthorized firmware exploits.

5. Newcomer Onboarding & Developer Setup Guide

Follow these steps to set up and run GridShield AI locally:

Step 1: Clone Repository & Install Dependencies

```
git clone https://github.com/eon-challenge/gridshield-ai.git
cd E.ON_Hackathon_Event
npm install
```

Step 2: Launch Development Server & Test Production Build

```
npm run dev # Starts local dev server at http://localhost:5173/
npx vite build # Verifies production bundle (100% clean success)
```

__GridShield AI Defensibility Report:__ Generated for E.ON Innovation Challenge 2026.